23.2 Exercises - Radioactive decay

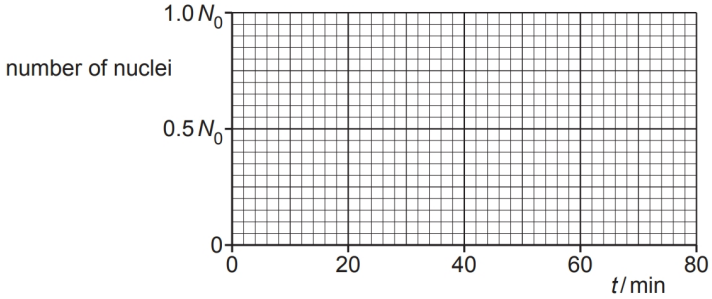
1. Carbon-11 is radioactive and decays by β+ emission to form boron-11. Carbon-11 has a half-life of 20 minutes. Boron-11 is stable.

a. Define half-life.

.....[1]

b. A sample contains N0 nuclei of carbon-11 and no other nuclei at time t = 0.

On Fig., sketch the variation with t of the number of nuclei of boron-11 in the sample. [3]



c.

i. Explain, with reference to the random nature of radioactive decay, why the activity of the carbon-11 sample in (b) decreases with time.

.....

.....[2]

ii. State, with reasons, whether a radiation detector placed near to the sample of carbon-11 indicates a measured count rate from the sample that is less than, the same as or greater than the activity of the sample.

.....

.....

.....[3]

2. Plutonium-238 ($^{238}_{94}Pu$) is unstable and undergoes alpha decay.

a. Complete the equation to show the decay of plutonium-238. [2]

$$^{238}_{94}Pu \rightarrow {}^{\cdots\cdots}_{\cdots\cdots}U + {}^{\cdots\cdots}_{\cdots\cdots}\alpha$$

b. The power source in a space probe contains 0.874kg of plutonium-238. Each nucleus of plutonium-238 that decays emits 5.59MeV of energy. The half-life of plutonium-238 is 87.7 years.

i. Calculate the initial number No of nuclei of plutonium-238 in the power source.

No = ... [1]

ii. Determine the initial activity of the source. Give a unit with your answer.

activity = ... unit ..... [2]

iii. Use your answer in (b)(ii) to determine the initial power output from the source due to the decay of plutonium-238.

power output = ... W [2]

iv. The space probe will continue to function until the power output from the plutonium in the source decreases to 65.3% of its initial value.

Calculate the time, in years, for which the space probe will function.

time = ... years [2]

c. An alternative power source uses energy generated from the radioactive decay of polonium-210. This isotope has a half-life of 0.378 years. The mass of the isotope needed for the same initial power output as in (b) is 3.37g.

advantage .....

disadvantage .....[2]

3.

a. Radioactive decay is both random and spontaneous.

State what is meant by:

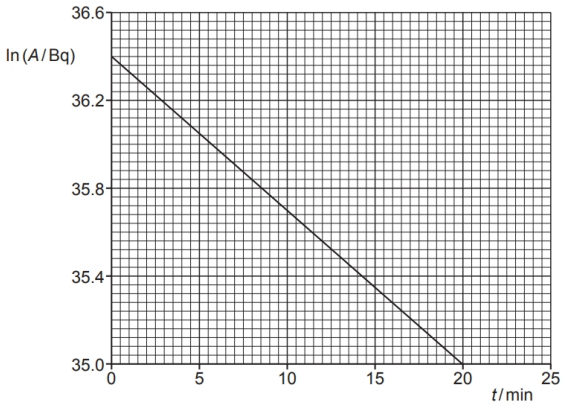
i. random

.....[1]

ii. spontaneous.

.....[1]

b. A sample of radioactive material contains atoms of an unstable nuclide X. The activity of the sample due to the atoms of X is A. The variation with time t of ln A is shown in Fig.



i. Use Fig. to determine the half-life, in minutes, of nuclide X.

half-life = ... min [3]

ii. At time t = 0, the mass of the atoms of X in the sample is 5.66×10^{-7} kg.

Determine the nucleon number of X.

nucleon number = ... [3]

4.

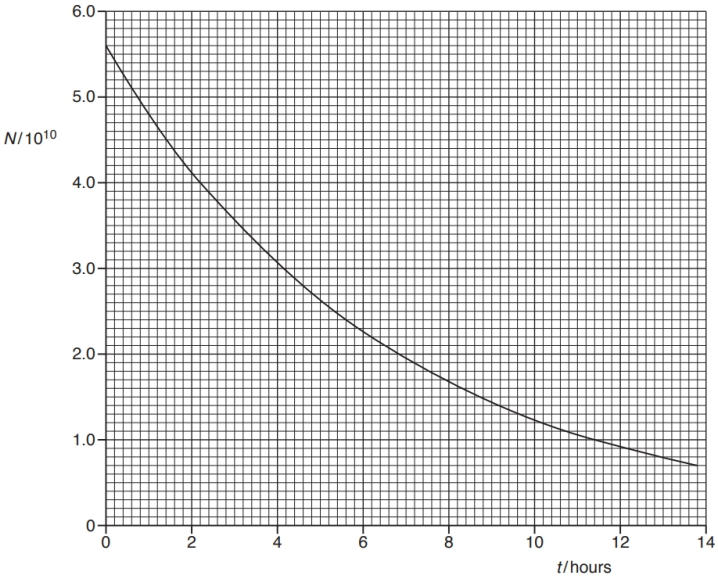
a. State what is meant by radioactive decay.

.....

.....

.....[3]

b. The variation with time t of the number N of undecayed nuclei in a sample of a radioactive isotope is shown in Fig.



i. Use the gradient of the line in Fig. to determine the activity, in Bq, of the sample at time t = 4.0 hours. Show your working.

activity = ... Bq [3]

ii. Use your answer in (i) to show that the decay constant λ of the isotope is approximately 4×10^{-5} s^{-1}. [2]

c. A sample of a different radioactive isotope has an initial activity of 4.6×10^{3}Bq. The sample must be stored safely until its activity is reduced to 1.0×10^{3}Bq.

The decay constant of the isotope is 5.5×10^{-7} s^{-1}. The decay products are not radioactive.

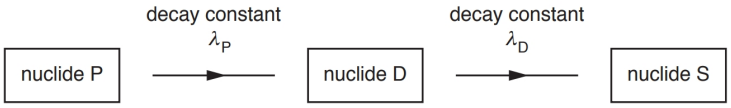
Calculate the minimum time, in days, for which the sample must be stored.

time = ... days [3]

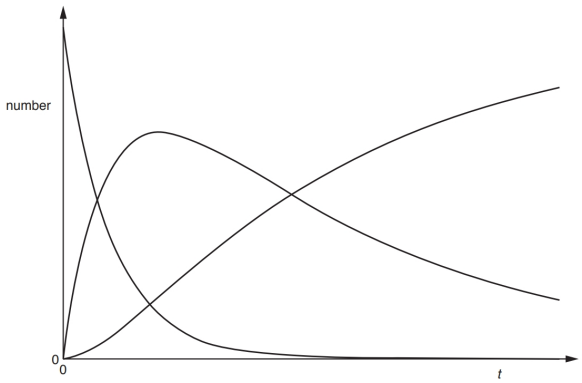
5.

a. An unstable nuclide P has decay constant λ_P and decays to form a nuclide D.

This nuclide D is unstable and decays with decay constant λ_D to form a stable nuclide S. The decay chain is illustrated in Fig.



The symbols P, D and S are not the nuclide symbols. Initially, a radioactive sample contains only nuclide P. The variation with time t of the number of nuclei of each of the three nuclides in the sample is shown in Fig.



i. On Fig., use the symbols P, D and S to identify the curve for each of the three nuclides. [2]

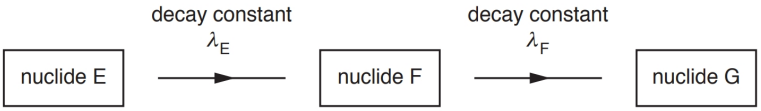
ii. The half-life of nuclide P is 60.0 minutes.

Calculate the decay constant λ_P, in s^{-1}, of this nuclide.

λ_P = ...s^{-1} [2]

b. In the decay chain shown in Fig. below, λ_P is approximately equal to $5\lambda_D$.

The decay chain of a different nuclide E is illustrated in Fig.



The decay constant λ_F of nuclide F is very much larger than the decay constant λ_E of nuclide E. By reference to the half-life of nuclide F, explain why the number of nuclei of nuclide F in the sample is always small.

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.....[2]